PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Heart Disease Prediction

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Heart Disease Prediction"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Mr. Gurpreet Singh. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Heart disease is one of the leading causes of mortality globally. Early prediction and detection of heart-related ailments can help reduce the number of fatalities and support timely medical treatment. In this project, we have developed a Machine Learning-based Heart Disease Prediction system using Random Forest Classifier, which predicts the presence of heart disease based on input parameters such as age, sex, cholesterol level, chest pain type, maximum heart rate, and others.

A web application was developed using Streamlit to provide an interactive user interface for end users. Additionally, data visualization and insights were implemented using Power BI to enhance interpretability. The system utilizes the ‘heart.csv’ dataset and achieves a high prediction accuracy with informative dashboards.

Keywords: Heart Disease Prediction, Machine Learning, Streamlit, Power BI, Random Forest, Data Visualization

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CHAPTER 1: INTRODUCTION

Heart disease, also known as cardiovascular disease (CVD), remains one of the most significant health challenges faced by societies worldwide. It includes a broad spectrum of disorders involving the heart and blood vessels, such as coronary artery disease, heart failure, arrhythmias, and valvular heart disease. These conditions generally result from the narrowing, blockage, or malfunctioning of blood vessels, leading to inadequate blood supply to the heart muscle and other vital organs. The consequences of untreated or late-diagnosed heart disease are often severe, ranging from angina (chest pain) and heart attacks (myocardial infarction) to stroke and sudden cardiac death.

According to the World Health Organization (WHO), cardiovascular diseases account for nearly 17.9 million deaths annually, representing about 32% of all global deaths, making them the leading cause of mortality worldwide. The burden of heart disease is disproportionately high in low- and middle-income countries due to limited access to healthcare, inadequate preventive measures, and rising prevalence of risk factors such as obesity, diabetes, hypertension, and sedentary lifestyles. Moreover, the economic impact of heart disease is profound, affecting workforce productivity and imposing significant costs on healthcare systems.

Early detection and intervention are critical to reducing the morbidity and mortality associated with heart disease. Conventional diagnostic methods include clinical assessments, laboratory tests, electrocardiograms (ECG), echocardiography, and angiography. However, these approaches can be time-consuming, costly, and sometimes invasive, limiting their widespread use, especially in resource-constrained settings. In recent years, advances in data science and artificial intelligence have paved the way for novel, cost-effective, and scalable solutions for disease prediction and diagnosis.

Machine learning (ML), a subset of artificial intelligence, enables computers to learn from data and identify complex patterns without explicit programming. ML models can process large volumes of heterogeneous health data—including demographic, clinical, biochemical, and imaging data—to predict disease risk and assist clinicians in making more accurate and timely decisions. These models have shown promise in various medical domains, including cancer detection, diabetic retinopathy screening, and cardiovascular risk stratification.

This project focuses on leveraging machine learning techniques to develop a predictive model for heart disease. The model is trained on a comprehensive dataset containing patient attributes such as age, sex, blood pressure, cholesterol levels, chest pain types, maximum heart rate achieved during exercise, and other relevant clinical parameters. The goal is to accurately classify patients as likely or unlikely to have heart disease, enabling early diagnosis and facilitating proactive healthcare measures.

A Random Forest Classifier is chosen as the primary predictive algorithm due to its

robustness, ability to handle both numerical and categorical variables, resistance to overfitting, and ease of interpretability through feature importance scores. To validate the model's effectiveness, it is trained and tested on a representative dataset, with performance evaluated through metrics such as accuracy, precision, recall, and F1-score.

To make this predictive capability accessible and practical, the model is embedded within a web-based application developed using Streamlit, a Python framework that simplifies building interactive data applications. This application allows users—including patients, doctors, and healthcare workers—to input relevant health data and instantly receive heart disease risk predictions through a friendly and intuitive interface.

Additionally, to provide further insights into the dataset and support data-driven decision making, an interactive dashboard is created using Microsoft Power BI. This dashboard visualizes key trends, distributions, and correlations among clinical parameters, shedding light on important risk factors and demographic influences on heart disease prevalence.

The motivation for this work arises from the urgent need for scalable, affordable, and accurate heart disease prediction tools, especially in regions where healthcare access is limited. By harnessing the power of machine learning and modern data visualization, this project aims to contribute towards reducing the global burden of cardiovascular diseases, improving early detection rates, and ultimately saving lives.

CHAPTER 2: SOFTWARE AND HARDWARE REQUIREMENTS

This chapter outlines the essential software and hardware components required to successfully develop, deploy, and operate the Heart Disease Prediction system. Proper selection and setup of these requirements are critical to ensuring smooth data processing, efficient model training, interactive application performance, and insightful data visualization.

2.1 Software Requirements

2.1.1 Programming Language

- Python 3.10 or higher: Python is chosen as the primary programming language due to its extensive libraries and frameworks tailored for data science, machine learning, and web application development. Its readable syntax and active community support accelerate development and troubleshooting.

2.1.2 Libraries and Frameworks

The following libraries form the backbone of the project's data processing, modeling, and visualization workflows:

- scikit-learn: This widely-used machine learning library offers efficient implementations of classification, regression, and clustering algorithms. The Random Forest classifier from scikit-learn is the core model used for heart disease prediction due to its robustness and interpretability.
- pandas: Provides data structures such as DataFrames, which facilitate efficient manipulation, cleaning, and preprocessing of structured data, including handling missing values, encoding categorical variables, and feature scaling.
- numpy: A fundamental package for scientific computing in Python, numpy supports efficient numerical operations on arrays and matrices, essential for mathematical computations during model training.
- matplotlib and seaborn: These libraries are utilized for exploratory data analysis (EDA) by generating a wide range of static, animated, and interactive visualizations. They assist in identifying data distributions, correlations, and outliers.

- Streamlit: A modern Python framework designed for rapid development of web applications, Streamlit enables building an interactive user interface for the predictive model with minimal coding overhead. It supports real-time data input, model inference, and display of results in an intuitive layout.

- Power BI Desktop: Microsoft's Power BI Desktop is employed to create rich, interactive dashboards for data visualization and business intelligence. It facilitates the transformation of raw data into meaningful insights through customizable charts, maps, and reports, aiding stakeholders in understanding key factors influencing heart disease.

2.1.3 Development Environment

- Integrated Development Environments (IDEs) such as Visual Studio Code, PyCharm, or Jupyter Notebook are recommended to write, test, and debug the Python code effectively.
- Package management tools like pip or conda simplify installation and updating of the required Python libraries, ensuring version compatibility.

2.2 Hardware Requirements

For effective execution of the various stages of this project—from data preprocessing to model training and application deployment—the following hardware specifications are recommended:

- Processor: A multi-core processor such as an Intel Core i5 or equivalent AMD Ryzen 5 series is recommended to handle the computational load involved in training machine learning models and running data-intensive applications. Multi-threading capabilities accelerate parallel processing within algorithms like Random Forest.
- RAM: A minimum of 8 GB of RAM is essential to manage the datasets efficiently during preprocessing, modeling, and visualization. Adequate memory ensures smooth execution without excessive swapping or lag, especially when working with larger datasets or running multiple applications concurrently.
- Storage: At least 1 GB of free disk space is required to install software packages, store datasets, trained model files, and temporary data generated during processing. Solid State Drives (SSD) are preferred over traditional Hard Disk Drives (HDD) for faster data access and overall system responsiveness.
- Operating System: The project can be developed and deployed on popular operating systems such as Windows 10 or higher, as well as Linux distributions

like Ubuntu. Compatibility with these platforms ensures flexibility in development environments and deployment targets.

2.3 Environment Setup and Configuration

Setting up a robust development environment is crucial to streamline the workflow and reduce setup time:

- Python Environment Setup: Creating isolated Python environments using tools like virtualenv or Conda environments is recommended to avoid dependency conflicts. Required libraries can be installed using pip install commands or Conda packages.
- Streamlit Deployment: The Streamlit application runs locally by default but can be deployed on cloud platforms such as Streamlit Cloud, Heroku, or AWS Elastic Beanstalk for broader accessibility. These platforms provide scalable infrastructure to host web applications with minimal configuration.
- Power BI Installation: Power BI Desktop is freely available from Microsoft's official website and requires a Windows environment. It offers a graphical user interface for importing datasets, creating reports, and publishing dashboards to the Power BI service for online access.
- Version Control: Using Git for version control is advisable to track code changes, collaborate with team members, and maintain project history.

2.4 Summary

The integration of these carefully chosen software and hardware components ensures that the Heart Disease Prediction system can be developed efficiently, deployed seamlessly, and used reliably. The combination of Python's powerful data science libraries, interactive visualization tools, and accessible hardware creates a conducive ecosystem for implementing advanced machine learning solutions tailored for healthcare applications.

CHAPTER 3: METHODOLOGY

The methodology adopted for the Heart Disease Prediction project is structured into several comprehensive stages. Each phase has been carefully designed to ensure the development of an accurate, reliable, and user-friendly prediction system, integrating machine learning with practical deployment and visualization tools.

3.1 Data Collection

The foundation of this project is the acquisition of a high-quality dataset representative of patients' health parameters related to heart disease. The dataset used, titled **'heart.csv'**, was sourced from the **UCI Machine Learning Repository**, a widely respected repository for machine learning datasets. This dataset includes detailed demographic and clinical information for a cohort of patients, encompassing attributes such as age, sex, blood pressure, cholesterol levels, chest pain types, electrocardiogram results, maximum heart rate achieved, and the presence or absence of heart disease.

Utilizing a well-established and publicly available dataset ensures reproducibility and reliability of the model, while providing sufficient feature variety to enable meaningful prediction. The data represents a realistic cross-section of patient health indicators relevant to cardiovascular risk assessment.

3.2 Data Preprocessing

Before feeding the data into machine learning algorithms, thorough preprocessing steps are necessary to prepare the dataset for effective analysis and modeling. The preprocessing pipeline consists of the following key steps:

- **Handling Missing Values**: The dataset was inspected for missing or inconsistent entries that could impair model training or prediction accuracy. Fortunately, the 'heart.csv' dataset contained no missing values; nevertheless, this verification step is crucial in real-world scenarios to maintain data integrity.

- **Encoding Categorical Variables**: Several features in the dataset, such as **chest pain type (cp)**, **thalassemia (thal)**, and **resting electrocardiogram results (restecg)**, are categorical in nature.

These non-numeric categories were transformed into numerical representations using encoding

techniques suitable for machine learning. Common methods include **one-hot encoding**, which creates binary indicator variables, and **label encoding**, which assigns ordinal integers. The choice of encoding preserves the categorical information while enabling algorithmic processing.

- **Feature Scaling**: Continuous variables like **cholesterol level (chol)**, **resting blood pressure (trestbps)**, and **maximum heart rate achieved (thalach)** were normalized or standardized. This scaling ensures that all features contribute proportionately during model training and improves convergence speed. Techniques such as **Min-Max Scaling** or **StandardScaler** (zero mean, unit variance) were applied based on data distribution.
- **Feature Selection**: To optimize model performance and reduce the risk of overfitting, relevant features were selected based on **statistical correlation analysis**, domain knowledge, and exploratory data analysis. Features highly correlated with the target variable (presence of heart disease) were retained, while redundant or less informative features were excluded, simplifying the model and enhancing interpretability.

3.3 Model Selection and Training

The core of the predictive system is the machine learning model responsible for classifying patient risk:

- A **Random Forest Classifier** was selected due to its **ensemble learning approach**, combining multiple decision trees to improve predictive accuracy and robustness against noisy data. It naturally handles both numerical and categorical variables and provides feature importance metrics, aiding interpretability.

- The dataset was partitioned into **training (80%)** and **testing (20%)** subsets. This split enables the evaluation of model generalization performance on unseen data.
- To fine-tune model performance, **hyperparameter optimization** was

conducted using **Grid Search** combined with **k-fold cross-validation** (commonly 5-fold). Parameters adjusted included:

  - Number of trees in the forest (n_estimators)
  - Maximum depth of each tree (max_depth)
  - Minimum samples required to split internal nodes (min_samples_split)
  - Minimum samples required at leaf nodes (min_samples_leaf)

This systematic search ensures selection of hyperparameters that balance bias and variance, resulting in optimal classification accuracy.

3.4 Model Evaluation

The trained model's effectiveness was rigorously evaluated using multiple performance metrics, each providing insights into different aspects of classification quality:

- **Accuracy**: Measures the overall proportion of correct predictions (both positive and negative) made by the model.
- **Precision**: Indicates the proportion of positive identifications that were actually correct, reflecting the model's ability to avoid false positives.
- **Recall (Sensitivity)**: Measures the proportion of actual positives correctly identified, important in medical diagnosis to minimize false negatives.
- **F1-Score**: The harmonic mean of precision and recall, providing a balanced measure that accounts for both false positives and false negatives.
- **Confusion Matrix**: A tabular visualization of true positives, true negatives, false positives, and false negatives that helps assess model performance in detail.

These metrics collectively guide the assessment of the model's reliability and clinical utility.

3.5 Application Development

To make the prediction model accessible to users without requiring programming

knowledge, an interactive web application was developed:

- Built using **Streamlit**, a Python-based framework facilitating rapid development of data-driven web applications.
- The application provides a clean and intuitive user interface with input widgets such as sliders, dropdowns, and numeric fields to enter patient health parameters.
- Upon submission, the app runs the trained Random Forest model in real-time, displaying the predicted likelihood of heart disease in a clear and user-friendly manner.
- Additional features include input validation to prevent erroneous entries and descriptive explanations of prediction outcomes to assist users in understanding their risk.
- The application supports deployment on local machines or cloud services for wider accessibility.

3.6 Data Visualization

To complement the predictive model, a robust data visualization component was developed using **Microsoft Power BI**:

- Interactive dashboards showcase critical insights such as demographic distributions (age, sex), prevalence of heart disease, and feature importance derived from the Random Forest model.
- Visual elements include bar charts, histograms, heatmaps, and scatter plots that allow users to explore data patterns dynamically.
- Filters and slicers enable tailored views based on user-selected criteria, enhancing data interpretability.

- These visual analytics aid healthcare professionals and stakeholders in understanding the factors influencing heart disease risk, supporting informed decision-making.

3.7 Testing and Validation

Comprehensive testing was undertaken to ensure the accuracy, robustness, and

usability of all system components:

- **Model Testing**: The predictive model was evaluated on the reserved test dataset to validate generalization. Multiple test scenarios, including edge cases, were used to check stability and performance.
- **Application Testing**: The Streamlit app underwent usability testing focusing on input validation, response time, and correct display of results. Tests ensured compatibility across devices and browsers.
- **Dashboard Testing**: Power BI dashboards were cross-verified against raw data for correctness, and interactive filters were tested for responsiveness and accuracy.
- **User Feedback**: Informal usability feedback was collected to improve interface clarity and user experience.

This iterative testing cycle ensured the final system is reliable, user-friendly, and suitable for deployment in practical healthcare environments.

CHAPTER 4: DATASET OVERVIEW

The dataset utilized for this project is the 'heart.csv' dataset from the UCI Machine Learning Repository. It contains health-related attributes of patients along with a target variable indicating the presence or absence of heart disease.

4.1 Dataset Features

The dataset comprises 14 attributes as described below:

Feature Name	Description	Data Type
age	Age of the patient in years	Continuous
sex	Sex (1 = male; 0 = female)	Categorical
cp	Chest pain type (values 0-3 indicating type)	Categorical
trestbps	Resting blood pressure (in mm Hg)	Continuous
chol	Serum cholesterol in mg/dl	Continuous
fbs	Fasting blood sugar > 120 mg/dl (1 = true; 0 = false)	Categorical
restecg	Resting electrocardiographic results (0,1,2)	Categorical
thalach	Maximum heart rate achieved	Continuous
exang	Exercise induced angina (1 = yes; 0 = no)	Categorical
oldpeak	ST depression induced by exercise relative to rest	Continuous
slope	Slope of the peak exercise ST segment	Categorical
ca	Number of major vessels (0-3) colored by fluoroscopy	Categorical

Feature Name	Description	Data Type
thal	Thalassemia (3 = normal; 6 = fixed defect; 7 = reversible defect)	Categorical
target	Presence of heart disease (1 = yes; 0 = no)	Categorical

4.2 Data Distribution and Insights

A thorough analysis of the dataset reveals important patterns and characteristics of the patient population and their health parameters, which are crucial for

effective model training and interpretation:

- **Age Distribution**: The ages of patients in the dataset span from 29 to 77 years, covering a broad adult demographic. The majority of patients fall within the 40 to 60 years age group, which aligns with the common onset period for cardiovascular diseases. Understanding this distribution helps the model focus on age-related risk factors and supports tailoring predictive insights for middle-aged and older adults.
- **Sex Distribution**: Analysis shows a higher prevalence of male patients compared to females in the dataset. This skew is consistent with epidemiological data that men have a higher risk of developing heart disease earlier in life compared to women. This imbalance is taken into account during model training to avoid biased predictions and ensure fair evaluation.
- **Chest Pain Types**: The dataset categorizes chest pain into four types: typical angina, atypical angina, non-anginal pain, and asymptomatic. These categories are distributed across the patient samples and are strongly indicative of cardiovascular health status. The inclusion of this categorical feature enriches the model's ability to differentiate between symptomatic and asymptomatic cases, contributing to prediction accuracy.
- **Serum Cholesterol Levels**: Cholesterol levels among patients show significant variability, ranging from low to very high values. Since cholesterol is a critical risk factor for heart disease, normalizing these continuous values is essential to prevent skewed influence on the model and to maintain numerical stability during training.

- **Other Feature Distributions**: Additional variables such as resting blood pressure, maximum heart rate achieved during exercise, and presence of exercise-induced angina also exhibit diverse ranges and patterns, each contributing unique predictive signals for the model.

Visual exploratory data analysis techniques such as histograms, box plots, and density plots were employed to visualize these distributions and identify any potential outliers or anomalies. These insights not only guided data preprocessing but also helped in feature selection and engineering.

4.3 Data Quality

The integrity and quality of the dataset directly impact the effectiveness of the predictive model. Key observations regarding data quality include:

- **Completeness**: The dataset contains no missing or null values, which significantly simplifies preprocessing steps and ensures that no imputation or data loss occurs. This completeness contributes to more reliable model training and evaluation.
- **Categorical Encoding Needs**: Several features in the dataset, including chest pain type, thalassemia, and resting electrocardiogram results, are categorical variables represented as non-numeric values or discrete codes. To prepare these features for machine learning algorithms, which typically require numerical inputs, appropriate encoding techniques such as label encoding and one-hot encoding were applied. This conversion preserves the informational content while enabling mathematical operations.
- **Data Consistency and Validity**: The dataset was checked for consistency in data types, value ranges, and logical coherence. For example, age values were verified to fall within humanly possible ranges, and categorical codes were cross-validated against documented categories to avoid mislabeling.
- **Exploratory Data Analysis (EDA)**: Comprehensive EDA was performed using graphical tools including:
    - **Histograms** to assess the frequency distribution of continuous variables.

- **Boxplots** to detect outliers and understand the spread of data.
- **Correlation Heatmaps** to evaluate linear relationships between features and identify multicollinearity, which can affect model performance.

This analysis informed decisions on feature selection, transformation, and scaling, ensuring that the data fed into the model is clean, representative, and conducive to accurate predictions.

CHAPTER 5: MACHINE LEARNING MODEL

This chapter elaborates on the entire machine learning pipeline implemented for the heart disease prediction system. It details the reasoning behind model choice, extensive training protocols, multifaceted evaluation criteria, feature relevance analysis, and comparative study with alternative models. The goal is to build a model that is not only accurate but also interpretable and clinically meaningful.

5.1 Model Selection

Selecting the appropriate machine learning algorithm is crucial to balance prediction accuracy, interpretability, and robustness—particularly in a sensitive domain like healthcare where decisions affect patient outcomes.

For this project, the **Random Forest Classifier** was chosen based on the following considerations:

- **Ensemble Learning Strength**: Random Forest aggregates predictions from multiple decision trees, reducing overfitting inherent in single trees. Each tree is trained on a bootstrap sample with randomized feature selection, encouraging diversity and stability in predictions.
- **Ability to Handle Mixed Data Types**: Patient data include both numerical features (e.g., cholesterol level, maximum heart rate) and categorical features (e.g., chest pain type, thalassemia). Random Forest naturally handles this heterogeneity without extensive preprocessing.

- **Robustness to Noise and Outliers**: Medical datasets often contain noisy or imbalanced data. Random Forest's voting mechanism minimizes the impact of anomalies on the final prediction.
- **Interpretability through Feature Importance**: Clinicians require explanations for model decisions. Random Forest provides feature importance metrics, enabling transparency and trust in the model.
- **Scalability and Efficiency**: Compared to other ensemble methods like boosting, Random Forest trains relatively quickly and can parallelize computations, making it practical for real-world deployment.

Given these advantages, Random Forest represents an ideal balance between predictive power and clinical interpretability.

5.2 Training Process

The training process was meticulously designed to optimize model generalization and robustness:

- **Data Partitioning**: The full dataset was randomly split into training and test subsets, with 80% reserved for training and 20% held out for final evaluation. This simulates future unseen data, ensuring unbiased performance estimates.
- **Hyperparameter Optimization**: Hyperparameters profoundly influence model behavior. The following parameters were tuned:
  - **Number of Trees (n_estimators)**: The number of decision trees in the forest controls the ensemble size. Larger forests typically improve stability but increase computational cost. Values tested ranged from 100 to 300 trees.
  - **Maximum Depth of Trees (max_depth)**: Controls tree complexity. Deeper trees can capture intricate patterns but risk overfitting. Optimal depths were explored to balance bias-variance trade-off.
  - **Minimum Samples Required to Split Nodes (min_samples_split)**: Affects node splitting sensitivity, preventing overly specific splits based on small sample sizes.

- **Grid Search with 5-Fold Cross-Validation**: A grid search systematically evaluated all combinations of hyperparameters across five folds. This involved:
    0. Splitting the training data into five subsets.
    1. Iteratively training on four folds and validating on the fifth.
    2. Averaging performance metrics across folds to assess stability.

This rigorous validation guards against overfitting and identifies hyperparameters that generalize well.

- **Final Model Training**: The best-performing hyperparameters from grid search were used to train the final Random Forest model on the entire training dataset, ready for evaluation on the test set.
- **Handling Class Imbalance**: If the dataset had an imbalance between heart disease positive and negative cases, techniques such as **class weighting** or

resampling would be applied. In this project, the dataset distribution was relatively balanced, simplifying training.

5.3 Evaluation Metrics

Comprehensive evaluation metrics were used to thoroughly assess the model's clinical and statistical performance:

- **Accuracy**: Measures the proportion of total correct predictions (both positive and negative). An accuracy of approximately **86%** indicates the model correctly classifies the majority of cases.
- **Precision**: The fraction of predicted positive cases that are true positives. High precision reduces unnecessary further testing or treatment caused by false alarms.
- **Recall (Sensitivity)**: The fraction of actual positive cases correctly detected. High recall is critical in healthcare to minimize missed diagnoses, which can lead to severe consequences.
- **F1-Score**: The harmonic mean of precision and recall, serving as a balanced metric when the costs of false positives and false negatives are both important.

- **Confusion Matrix**: Breaks down prediction results into true positives, true negatives, false positives, and false negatives. This provides detailed insight into error types, guiding clinical risk assessment.
- **ROC Curve and AUC (Area Under Curve)**: Although not explicitly mentioned, plotting the ROC curve and calculating AUC are valuable for visualizing the trade-off between sensitivity and specificity across thresholds.
- **Statistical Significance Testing**: Where applicable, performance differences were assessed for significance to ensure improvements are meaningful and not due to chance.

5.4 Feature Importance

Understanding which features drive the model's decisions is essential for clinical adoption and trust:

- The Random Forest model computes **feature importance scores** based on the average decrease in node impurity contributed by each feature across all trees.
- Top predictors identified include:
  - **Chest Pain Type (cp)**: Captures clinical symptoms strongly correlated with heart disease severity.
  - **Maximum Heart Rate Achieved (thalach)**: Indicates cardiovascular fitness and stress response.
  - **Serum Cholesterol Level (chol)**: Elevated cholesterol is a primary risk factor for plaque formation in arteries.
- These align well with medical expertise and literature, reinforcing model validity.
- Feature importance analysis also informs possible feature reduction, simplifying future models and reducing data collection burden.

5.5 Model Comparison

Alternative machine learning algorithms were implemented and benchmarked to validate the suitability of Random Forest:

- **Logistic Regression**: A linear model widely used in clinical risk prediction. Its simplicity provides easy interpretability but may lack the capacity to model complex nonlinear interactions between variables.
- **Support Vector Machines (SVM)**: A powerful classifier capable of handling nonlinear boundaries through kernel functions. However, SVMs can be sensitive to parameter settings and less interpretable, posing challenges in medical contexts.
- **K-Nearest Neighbors (KNN)** and **Decision Trees** were also briefly evaluated but showed inferior performance or higher susceptibility to overfitting.

The Random Forest consistently outperformed these methods in cross-validation accuracy, robustness to noisy features, and interpretability. Its ensemble approach effectively captured the nonlinear relationships inherent in cardiovascular risk factors.

5.6 Summary and Clinical Implications

The Random Forest model developed in this project exhibits high predictive accuracy, balanced precision and recall, and strong interpretability. Its alignment with known medical risk factors enhances trustworthiness and potential integration into clinical workflows.

By providing reliable heart disease risk assessments based on routinely collected clinical data, this model can support early diagnosis, personalized treatment planning, and improved patient outcomes. Furthermore, the model's ability to highlight key predictive features empowers clinicians to focus on critical health indicators.

Future work may explore integrating additional data modalities (e.g., imaging, genetic data), expanding datasets for broader generalizability, and incorporating explainable AI techniques to further enhance transparency.

CHAPTER 6: STREAMLIT WEB APPLICATION

6.1 Overview

The project includes the development of a user-friendly web application using **Streamlit**, a Python framework designed for building interactive data apps with minimal effort.

6.2 Application Features

- **Input Interface:**
  The app provides intuitive input widgets such as sliders, dropdowns, and numeric input fields for users to enter their health parameters (age, sex, chest pain type, cholesterol, etc.).
- **Real-Time Prediction:**
  Upon submitting the inputs, the Random Forest model predicts the likelihood of heart disease and displays the result instantly.
- **Result Explanation:**
  The app shows a simple explanation of the prediction outcome for better user understanding.

6.3 User Experience

- Designed to be minimalistic and responsive to ensure usability on different devices.

- Input validation ensures only valid data is accepted.
- Users receive clear feedback, including warnings or recommendations based on their inputs.

6.4 Deployment

- The Streamlit app runs locally but can be easily deployed on cloud platforms such as **Heroku**, **Streamlit Cloud**, or **AWS** for broader access.
- The lightweight nature of the app makes it suitable for low-resource environments.

6.5 Screenshots and Interface

- *Figure 3* in the report shows screenshots of the web application interface, demonstrating the input form and prediction output.

CHAPTER 7: POWER BI DASHBOARD

7.1 Purpose and Overview

The Power BI dashboard was developed to provide visual analytics and insights into the heart disease dataset. It helps users and healthcare professionals better understand key patterns and risk factors.

7.2Dashboard Components

- **Gender-wise Distribution:**
  Visualizes the prevalence of heart disease among males and females, highlighting any significant disparities.
- **Chest Pain Types:**
  Displays the distribution of different chest pain types and their correlation with heart disease presence.
- **Feature Correlation Heatmap:**
  Presents correlations between various features such as age, cholesterol levels, blood pressure, and target variable.
- **Age Distribution:**
  Shows histogram and boxplot representations of patient age ranges, indicating high-risk age groups.

7.3 Insights from Visualizations

- Identifies which features have stronger associations with heart disease.
- Helps in spotting outliers or data imbalances that may affect modeling.
- Supports data-driven decision-making and awareness building.

7.4 Dashboard Usability

- Interactive filters enable dynamic data exploration by feature or patient demographics.
- The dashboard is designed with clarity and simplicity for easy interpretation.

7.5 Future Enhancements

- Incorporation of real-time data streaming for live updates.
- Addition of predictive analytics widgets integrated with the model.
- User customization options for deeper exploratory analysis.

CHAPTER 8: TESTING AND EVALUATION

8.1 Model Testing

- The Random Forest model was tested on the 20% reserved test dataset.
- Achieved an **accuracy of approximately 86%**, indicating robust predictive performance.
- Additional metrics like precision, recall, and F1-score were computed to ensure balanced classification.

8.2 Application Testing

- The Streamlit web app was manually tested for usability, input validation, and correct prediction output.
- Various input scenarios were evaluated, including edge cases and invalid data entries, to ensure robustness.
- The app provided consistent and accurate prediction feedback in all cases.

8.3 Dashboard Testing

- Power BI dashboards were verified for data correctness and dynamic filtering capabilities.

- Cross-checked visualizations with raw dataset statistics to confirm accuracy.
- Ensured dashboard performance was smooth and without lag on typical hardware.

8.4 Limitations Observed During Testing

- The static dataset limits real-time predictive capabilities.
- The current Streamlit app does not support multi-user authentication or data storage.
- Power BI dashboards require manual refresh for new data inclusion.

8.5 Summary

The testing phase validated the reliability of the machine learning model, usability of the web application, and effectiveness of the visualization dashboards, establishing the system as a viable tool for heart disease prediction and analysis.

CHAPTER 9: LIMITATIONS

9.1 Dataset Limitations

- The dataset used for training and testing is relatively small and may not represent the full diversity of populations or heart disease presentations globally.
- It lacks real-time patient data and longitudinal records which can improve prediction accuracy.

9.2 Model Limitations

- The model performance depends heavily on the dataset quality; bias in data may lead to biased predictions.
- Some subtle risk factors may not be captured or represented adequately.
- The Random Forest model, while robust, may not be optimal for all possible input feature combinations.

9.3 Application Limitations

- The Streamlit application lacks user authentication and data privacy mechanisms, limiting clinical deployment.

- No facility for users to save or track their historical predictions.

9.4 Dashboard Limitations

- Power BI dashboards rely on manual data refresh, limiting the ability to monitor live changes or trends.
- Visualization features are basic and can be enhanced with more complex analytics and drill-downs.

9.5 General Limitations

- The system does not replace professional medical diagnosis and should be used as a supplementary tool.
- Ethical considerations around data usage and patient privacy need careful management.

CHAPTER 10: CONCLUSION

- This project successfully developed a machine learning-based system to predict heart disease using patient health data. The use of a Random Forest Classifier provided robust and accurate predictions, achieving an accuracy of approximately 86%.
- The integration of the model into a user-friendly Streamlit web application enables individuals and healthcare providers to assess heart disease risk conveniently. Complementary Power BI dashboards offer valuable insights into data patterns and risk factors.
- Despite certain limitations in dataset size, real-time capabilities, and application features, this project demonstrates the potential of AI-assisted healthcare solutions to improve early disease detection and patient outcomes.
- Future work can focus on incorporating larger, real-time datasets, enhancing app security and user experience, and expanding visualization and analytics features to make the system more comprehensive and clinically relevant.

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